

Agentic fMRIPrep Error Report

Summary of errors encountered during smoke-test and real toy-data runs, plus how the pipeline or operator resolved them.

Current Assessment

- The agent flow is working well for the tested toy-data pipeline: planning, execution, diagnosis, recovery, rerun, and QA all completed on the final real run.
- It is not accurate to call the system perfect yet. It still uses deterministic recovery rules and a few configured defaults such as fallback readout time and optional convenience flags.
- The hardcoded incorrect output-path assumption was removed; QA now discovers the real derivative layout and writes structured results to `agentic_results/sub-01/`.

Encountered Errors

- Synthetic smoke-test OOM failure: the mock docker shim forced an out-of-memory style exit 137. Diagnosis labeled it as memory pressure, recovery added `--low-mem` and reduced `nprocs` to 1, and the second attempt succeeded. This proved the orchestration loop.
- Real environment failure: Docker daemon unavailable or inaccessible. Diagnosis correctly identified a Docker daemon/permission problem. This was not fixed by code; it required starting Docker and making the socket reachable from the shell.
- Real fMRIPrep failure: missing readout timing metadata for the BOLD files. Initially the pipeline failed during SDC/fmap setup. Recovery was upgraded to add `--fallback-total-readout-time 0.05`, and config generation was further improved to add that flag proactively when metadata inspection shows it is needed.
- Real QA-path failure: the vision step originally assumed outputs would live under `output_dir/fmriprep/sub-01/`. The successful run actually wrote derivatives under `output_dir/sub-01/`. The QA code was changed to resolve the real subject derivative directory from the output tree instead of assuming one fixed layout.
- Real reporting issue: QA originally exposed an arbitrary Quality Score formula. That was replaced with concrete metrics: raw nonzero voxels, preprocessed nonzero voxels, brain-mask nonzero voxels, retention ratio, skull-strip reduction, MNI output presence, and a simple QA Summary.

How The Model Fixed Them

- Diagnosis logic now chooses one primary cause with evidence instead of emitting a noisy stack of loosely matching regex labels.
- Recovery logic now retries only when it produced a real command change; otherwise it stops instead of looping on the same command.
- Config generation now inspects BOLD sidecar metadata before launch and can proactively include `--fallback-total-readout-time 0.05` for non-anat runs when `TotalReadoutTime` or `EffectiveEchoSpacing`-based timing cannot be inferred.
- QA results are now printed in the final report and saved to `qa_report.txt` and `qa_metrics.json` under the run output.

Latest Successful Real Run

- Run: `tmp/real_toy_outputs_retry4`
- Attempts: 1
- QA Summary: PASS
- Raw voxels: 262,144; preprocessed voxels: 261,502; brain-mask voxels: 258,048
- Brain-mask retention ratio: 98.4%; skull-strip reduction: 1.6%
- Normalization: MNI-space preprocessed output file detected

Result Files

- QA report: `tmp/real_toy_outputs_retry4/agentic_results/sub-01/qa_report.txt`
- QA metrics: `tmp/real_toy_outputs_retry4/agentic_results/sub-01/qa_metrics.json`
- Run report: `tmp/real_toy_outputs_retry4/sub-01.html`